

Consumption Tilt Across Periods with Different Equivalence Scales

Divorce branch – modeling note

August 17, 2026

1 Setup

Two periods with equivalence scales $s_1 = \sqrt{\text{hh_size}_1}$, $s_2 = \sqrt{\text{hh_size}_2}$ and utility $u(C_t) = ((C_t/s_t)^{1-\mu} - 1)/(1-\mu)$ (no household-size multiplier in front, per `utility_functions_add.py`). Maximizing $u(C_1) + \beta u(C_2)$ subject to $C_1 + C_2/R = W$ gives the Euler equation

$$\frac{C_2}{C_1} = (\beta R)^{1/\mu} \left(\frac{s_1}{s_2} \right)^{(1-\mu)/\mu}. \quad (1)$$

For comparison, with the $\text{hh_size}_t = s_t^2$ multiplier back in front ($\tilde{u}(C_t) = \text{hh_size}_t \cdot u(C_t)$, the pre-fix version), the same steps give

$$\frac{C_2}{C_1} = (\beta R)^{1/\mu} \left(\frac{s_1}{s_2} \right)^{-(\mu+1)/\mu}. \quad (2)$$

2 Result

The sign of the exponent on (s_1/s_2) determines which period the tilt favors. Without the multiplier the sign depends on μ ; with it, the exponent $-(\mu+1)/\mu$ is negative for every $\mu > 0$, so it *never* flips – it always favors the larger household. Table 1 gives the direction for each regime, independent of which period happens to be larger.

μ	Pre-fix (with hh_size_t multiplier)	Post-fix (no multiplier)	example ($s_1 > s_2$)
$\mu > 1$	larger scale	larger scale	period 1 (partnered)
$\mu = 1$	larger scale	no tilt, $C_2/C_1 = \beta R$	period 1 (partnered)
$\mu < 1$	larger scale	smaller scale	period 2 (alone)

Table 1: Direction of the equivalence-scale tilt in the Euler equation: which period’s total consumption gets the extra share, beyond what βR alone prescribes.

The estimated model has $\mu_{\text{low}} = \mu_{\text{high}} = 0.8 < 1$ (`est_params_alg1_sparse.pkl`). Pre-fix, consumption always tilted toward the *larger*-household period, regardless of μ . Post-fix (current code), it tilts toward the *smaller*-household period – toward being alone – because removing the multiplier is what made the tilt direction μ -dependent, and the estimated $\mu < 1$ lands on the reversed side.